

CARLOS MARTINEZ LOPEZ

carlos.martinez@gmail.com

+34 691 445 382

<https://www.linkedin.com/in/carlos-martinez-lopez-ml>

<https://github.com/carlosmartinez>

PROFESSIONAL SUMMARY

Machine Learning Engineer with 5 years of experience building and deploying production ML systems for startups and scale-ups. Specialized in computer vision and NLP applications, with a track record of reducing model inference time by 40% and improving model accuracy through careful feature engineering. I enjoy working with cross-functional teams and translating complex technical concepts into actionable insights for non-technical stakeholders.

WORK EXPERIENCE

Senior Machine Learning Engineer

VisionAI Technologies

<https://visionai.io>

Madrid, Spain

March 2022 - Present

Developed and deployed a real-time object detection system using YOLOv8 that reduced false positives by 35% and increased inference speed to 45 FPS on edge devices.

Led the migration of legacy TensorFlow models to PyTorch, improving team productivity and reducing training time by 50% across all projects.

Designed and implemented an automated ML pipeline using MLflow and Airflow that decreased model retraining cycles from weekly to daily without increasing infrastructure costs.

Mentored 2 junior engineers on best practices in model evaluation, data validation, and production deployment strategies.

Machine Learning Engineer

DataDrive Solutions

<https://datadrive.co>

Madrid, Spain

June 2020 - February 2022

Built end-to-end NLP pipeline for customer sentiment analysis, processing 2 million reviews monthly and achieving 88% accuracy with BERT-based models.

Conducted comprehensive A/B testing framework that enabled the company to validate 15+ model variations monthly and select optimal configurations based on business metrics.

Optimized database queries and implemented caching strategies that reduced API response times from 800ms to 120ms for recommendation system endpoints.

Collaborated with product team to define success metrics and worked backward from business objectives to design appropriate ML solutions.

Junior Machine Learning Engineer

InnovateTech Systems

<https://innovatetech.net>

Madrid, Spain

January 2019 - May 2020

Preprocessed and cleaned large-scale tabular datasets using pandas and developed feature engineering pipelines that improved model performance by 12% on average.

Implemented classification models using scikit-learn and XGBoost for customer churn prediction, achieving AUC-ROC of 0.87 on test set.

Created data visualization dashboards using Matplotlib and Seaborn to communicate model insights to stakeholders and identify data quality issues.

Participated in code reviews and contributed to internal documentation, establishing best practices for model versioning and experiment tracking.

EDUCATION

Master of Science in Machine Learning and Data Science

Universidad Autonoma de Madrid

Madrid, Spain

2017 - 2019

GPA 3.7/4.0

Bachelor of Science in Computer Science

Universidad Autonoma de Madrid

Madrid, Spain

2013 - 2017

GPA 3.5/4.0

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, Bash

Machine Learning Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM

Data Tools: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter

ML Operations: MLflow, Airflow, DVC, Git, Docker, Kubernetes

Cloud Platforms: AWS SageMaker, Google Cloud AI Platform, Azure ML

Specializations: Computer Vision, Natural Language Processing, Time Series Forecasting, Model Optimization, Feature Engineering

GITHUB REPOSITORIES

customer-sentiment-nlp: End-to-end NLP pipeline for sentiment analysis using BERT and transformers, 240 stars

yolo-edge-optimizer: Framework for optimizing YOLOv8 models for edge deployment, achieving 3x speedup, 180 stars

xgboost-feature-engineering: Automated feature generation and selection toolkit for tabular data, 150 stars

time-series-anomaly-detection: LSTM-based anomaly detection system with interactive visualizations, 95 stars

LANGUAGES

Spanish: Native

English: Fluent

Portuguese: Conversational